

findings

CodeGraph Daemon #850 Watchdog Stability — Findings

Field	Value
Date	2026-06-22
Symptom	Recurring “Main thread unresponsive ~60s — killing wedged process (#850)”
Verdict	Index massively oversized because .codegraph/config.json is INERT
Query reliability	Reliable (sub-ms reads); only intermittent fail inside the ~60s wedge window

Root cause (FACT, evidence-backed)

CodeGraph v1.0.1 is **zero-config** and excludes **only via .gitignore** (official docs, accessed 2026-06-22). The repo’s .codegraph/config.json (legacy schema, dated Jun 3) is **no longer read** — it is inert.

Proof — the live DB tracks .gitignore, NOT config.json: - config.json excludes cli_agents/**, yet **36,099 cli_agents files are indexed** (git check-ignore cli_agents → NOT IGNORED; root .gitignore still lists the pre-CONST-052 name Example_Projects/, never the renamed cli_agents/). - cli_agents_resources/ (514) and github_pages_website/ (9) also “excluded” in config but indexed (not in .gitignore). - dependencies/ {LLama_CPP,Ollama,HuggingFace_Hub} = 0 indexed only because they are uninitialized (empty) submodules — NOT because the config worked.

Result: **96,674 files, 1,664,519 nodes, 3,829,858 edges → 4.38 GB DB** (submodules/ 57,456 + cli_agents/ 36,099 + nested vendored submodule trees with generated *.gen.ts/worker-configuration.d.ts). Each watcher re-walk/sync over that tree blocks the main thread >60s on an 18 GB host under heavy memory pressure (vm_stat: ~87 MB free, ~7.6 GB compressed) → the **#850 watchdog SIGKILLS and restarts** in a loop. Compounding: **two daemons run** (one with --path, one without), doubling load.

CRITICAL corollary — `codegraph_validate.sh` is a **PASS-bluff**

`codegraph_validate.sh` returned **18/18 PASS** by checking `config.json`'s include/exclude lists — but the tool ignores that file. The validator certifies an exclusion that does not happen. Third-party `cli_agents/**` IS indexed despite the “`cli_agents/** is excluded`” line. This is a §11.4 PASS-bluff at the validation layer and partially invalidates the earlier §11.4.79 “own-org included / third-party excluded” conclusion (own-org IS included — confirmed by live explore — but third-party is NOT actually excluded).

Timeline

Current `daemon.log` (40 lines, one window): **9 watchdog kills**, 13 restarts, 4 clean idle-timeouts. Lines 1–20: tight start → wedge → kill → restart loop (7× back-to-back, ~1–2 min apart); calm middle (3 idle shutdowns); then 3 more kills. Every kill = main thread blocked ≥60s (the node `-e` sidecar's 60000 ms timer, confirmed via `ps`).

Query reliability

Reliable. All 4 live `codegraph_explore` calls returned full correct results incl. own-org submodule symbols (`submodules/helix_qa`, `submodules/database`, `submodules/llms_verifier`, `submodules/containers`). SQLite reads sub-ms even at 4.38 GB; the stall is in the sync/index path, not the read path. Index grew 96,674 → 96,676 mid-session (watcher actively re-walking). Caveat: a read landing in the ~60s wedge can intermittently fail and need a retry.

Recommended mitigation (recommendation only — NOT applied)

- **A. Fix scope via the mechanism CodeGraph honors (`.gitignore` / `.git/info/exclude`), not `config.json`.** Exclude `cli_agents/`, `cli_agents_resources/`, `github_pages_website/`, nested submodules/`**/submodules/**` + `submodules/**/cli_agents/**`, and generated `**/*.gen.ts` / `**/worker-configuration.d.ts`. Since `cli_agents` must stay git-tracked, prefer non-tracking `.git/info/exclude` over editing committed `.gitignore`; confirm mechanism with operator.
- **B. Stop daemon(s), delete `codegraph.db*`, re-run `codegraph index` — DB should shrink to a fraction; stalls disappear.**
- **C. Reconcile MCP config to ONE serve `--mcp --path ... daemon`, not two.**

- **D.** Watchdog tuning (CODEGRAPH_WATCHDOG_TIMEOUT_MS / CODEGRAPH_NO_WATCHDOG=1) is a stop-gap only — disabling reintroduces the 100%-CPU hang #850 prevents; fix scope first.
- **E. Constitution-level (§11.4.78/§11.4.79):** document exclude source-of-truth = .gitignore NOT config.json; add to codegraph_validate.sh: (1) index-size budget
 - SELECT COUNT(*) FROM files WHERE path LIKE '<excluded>/%' MUST be 0 (non-zero = inert exclude = release-blocker, mirroring §11.4.79's “index that lies is a PASS-bluff”); (2) daemon-health probe that FAILs if #850 fired in recent log;
- 1. one-daemon-per-path invariant; (4) CONST-052 renames MUST cascade into ignore files in the same change.

Sources (verified 2026-06-22)

- <https://github.com/colbymchenry/codegraph/releases>
- <https://colbymchenry.github.io/codegraph/getting-started/configuration/>
- <https://colbymchenry.github.io/codegraph/guides/indexing/>
- <https://colbymchenry.github.io/codegraph/troubleshooting/>